

Break Your Own Site — Hardening & Diligence Report

Project: EcoHive Kenya Ltd.

Track: General AI Fluency (Week 7)

Live Application URL: <https://ais-pre-vegkej4u2lmpqczdp2pstl-122122115705.europe-west2.run.app>

Date: September 2026

1. Executive Summary

This document records the results of deliberately testing, breaking, and hardening the **EcoHive Kenya Ltd.** digital platform. We moved beyond the happy path by subjecting the forms, API endpoints, navigation states, and rendering pipelines to empty inputs, garbage data, malformed payloads, rapid double-submits, unhandled routes, and low-power rendering constraints.

Every finding was triaged into **Fix-Now** (resolved immediately) or **Known Limitation** (documented and architecturally bounded).

2. The Honest "Where It Breaks" Triage

#	Test Scenario / Breaking Attempt	Pre-Hardening Behavior	Triage	Resolution / Fix Implemented
1	Empty & Whitespace Form Submissions	Submitting " " or leaving inputs blank passed basic client checks and created empty database leads ("Thank you, !").	Fix-Now	Implemented backend & frontend string sanitization, minimum length constraints (≥ 2 chars for name, ≥ 5 for message), and strict RFC-compliant email regex validation.
2	Garbage / Malformed JSON Payloads	Sending <code>{malformed}</code> to <code>/api/*</code> caused the Express JSON parser to crash with an unhandled HTML <code>SyntaxError</code> stack trace.	Fix-Now	Added Express error-handling middleware that intercepts JSON <code>SyntaxError</code> and returns a clean, secure HTTP 400 JSON payload: <code>{ "error": "Malformed JSON payload provided." }</code> .
3	Rapid Double-Submitting Forms	Double-clicking or spamming submit sent duplicate simultaneous requests, creating duplicate lead entries.	Fix-Now	Enforced client-side submit lock (<code>submitting</code> state disables button and displays loading spinner) and clamped numeric inputs (<code>hiveQuantity</code> bounded to <code>1-1000</code>).
4	Querying Non-Existent API	Unmatched <code>/api/*</code> requests fell through to the Vite SPA fallback, returning	Fix-Now	Added an explicit <code>app.all('/api/*', ...)</code> catch-all returning structured

#	Test Scenario / Breaking Attempt	Pre-Hardening Behavior	Triage	Resolution / Fix Implemented
	Routes (GET /api/unknown)	HTTP 200 with raw HTML instead of a JSON REST error.		HTTP 404 JSON: { "error": "API route not found: METHOD PATH" }.
5	Empty Search & Catalog Zero-States	Searching for non-existent items or empty filter categories rendered a blank, dead grid without user feedback.	Fix-Now	Built an interactive product search bar with an informative zero-results empty-state card and a one-click "Reset Search & Show All" button.
6	Route Transitions Retaining Prior Scroll Position	Navigating between views while scrolled down preserved the scroll offset, causing new views to load mid-page.	Fix-Now	Added an automated <code>useEffect</code> hook in <code>App.tsx</code> executing <code>window.scrollTo({ top: 0, behavior: 'instant' })</code> whenever <code>currentPage</code> changes.
7	Social Link Sharing & Findability	Missing OpenGraph/Twitter image tags produced blank link previews on WhatsApp, LinkedIn, and Twitter.	Fix-Now	Added 1200x630 <code>og:image</code> , <code>twitter:image</code> , canonical URL tags, author tags, and Schema.org <code>Organization</code> JSON-LD structured data in <code>index.html</code> .
8	Hardware WebGL Disabled or Low-Power Device	Legacy smartphones or browsers with hardware acceleration turned off failed to initialize the Three.js 3D viewport.	Known Limitation	Handled with an integrated 2D High-Resolution Render mode featuring interactive pinned hotspots, allowing 100% feature parity on devices without WebGL.
9	Real-world Apiary MQTT Stream vs. Simulation	Telemetry currently streams from the Express server rather than live physical satellite hardware in Baringo/Nakuru.	Known Limitation	The architecture is decoupled: <code>/api/iot-telemetry</code> provides a drop-in contract ready to point to physical GSM hardware brokers without frontend refactoring.

3. Speed & Performance Benchmarks

All response times measured on the live deployment:

Endpoint / Asset	Latency	Status
Root Document (/)	~53 ms	HTTP 200
IoT Telemetry (/api/iot-telemetry)	~4.0 ms	HTTP 200
Health Check (/api/health)	~3.2 ms	HTTP 200
API Error Handling (/api/404)	~3.5 ms	HTTP 404 JSON

Endpoint / Asset	Latency	Status
3D Model Payload	0 KB (Procedural)	WebGL 60 FPS

4. How to Verify

- 1. **Empty Form Test:** Navigate to **About & Contact**, leave fields blank or type spaces, and click *Send Inquiry*. Notice specific validation errors.
- 2. **Double-Click Test:** Enter valid info and click submit rapidly; the button locks immediately to prevent double submissions.
- 3. **Search Zero-State:** Go to **Products & Honey**, type "xyz123" in the search bar, and verify the friendly empty state and reset button.
- 4. **WebGL Fallback:** Go to **The Smart Hive** and click the "2D Hotspots" toggle to test the non-WebGL experience.